

Analysis of the "Distortion" Phenomenon in Retarded Potentials

1 Definitions

- $\rho_{(field)}(\mathbf{r}', \mathbf{r}, t)$ is the charge density at the **field point** $\mathbf{r}'$ at the **retarded time**, i.e., the charge density from the field's perspective.
- $\rho_{source}(\mathbf{r}', t)$ is the charge density at the **source point** $\mathbf{r}'$ at the **source time**, i.e., the charge density from the source's perspective.

2 Definition of Distortion

After fixing two of the three quantities $(\mathbf{r}, \mathbf{r}', t)$, the distribution of the field-perspective ρ (which then depends solely on the third quantity) in the space of that third quantity, under temporal translation, differs from the distribution of the source-perspective $\rho_{source}(\mathbf{r}', t)$.

3 Case Analysis

1. Example 1: Fixed Field Point and Source Point

Fixed $\mathbf{r}_0$ and $\mathbf{r}'_0$. Here, the field-perspective ρ (depending only on t) equals the source-perspective $\rho_{source}(\mathbf{r}', t - R/c) = \rho_{source}(\mathbf{r}'_0, t - R/c)$, because $R = |\mathbf{r}_0 - \mathbf{r}'_0|$ is constant here, and ρ_{source} depends only on t .

Conclusion: On the one-dimensional time axis of the field perspective, the structure of constant values consists of points. After a temporal translation of R/c , it coincides with the time axis of the source perspective. Hence, there is **no distortion**.

2. Example 2: Fixed Field Point and Observation Time

Fixed $\mathbf{r}_0$ and t_0 . The field-perspective ρ depends only on $\mathbf{r}'$. On a circle (2D) or sphere (3D) centered at $\mathbf{r}_0$ (a contour/surface of constant R), the value of $\rho(\mathbf{r}')$ for all points corresponds to the value of the source-perspective $\rho_{source}(\mathbf{r}', t_0 - R/c)$ at that point at time $t_0 - R/c$.

However, R is a variable here, and unless all source charges are equidistant from $\mathbf{r}_0$ (which is unphysical because point charges in electrodynamics are not mathematical points but have volume distribution, thus having distribution towards $\mathbf{r}_0$), the field-perspective $\rho(\mathbf{r}', t_0)$ and the source-perspective $\rho_{\text{source}}(\mathbf{r}', t_0)$ cannot be related by a constant temporal translation (the translation factor R/c is not constant).

Conclusion: Therefore, it is **distorted**.

3. **Example 3: Fixed Source Point and Observation Time**

Fixed $\mathbf{r}'_0$ and t_0 . The source-perspective $\rho_{\text{source}}(\mathbf{r}', t_0) = \rho_{\text{source}}(\mathbf{r}'_0, t_0)$ is a constant value. However, the field-perspective ρ depends only on $\mathbf{r}$. On a circle/sphere of radius R centered at $\mathbf{r}'_0$, the value of ρ for all points $\mathbf{r}$ corresponds to the value of the source-perspective ρ_{source} at that point at the time $t_0 - R/c$.

However, R is a variable here, and unless all field points are equidistant from $\mathbf{r}'_0$, the field-perspective ρ and the source-perspective ρ_{source} cannot be related by a constant temporal translation (the translation factor R/c is similarly not constant).

Conclusion: Therefore, this is also **distorted**.